**Submitted by:** Tanmay Kishor

**Program:** Data Science and Machine Learning Summer Internship

**GitHub Repository Link:** https://github.com/tanmay-kishor/upskillcampus/blob/main/Project5_Crop_Weed_Detection.ipynb

---

## 1. Introduction & Objective

In modern precision agriculture, automated and rapid identification of crops versus weeds is critical for targeted herbicide application, reducing chemical usage, and improving crop yield. The objective of this project is to build, train, and evaluate a state-of-the-art computer vision model using **Yolov8 (Nano)** to accurately detect and classify crops and weeds from agricultural field images.

## 2. Dataset & Preprocessing

- **Source Dataset:** Agricultural crop and weed image database.
- **Dataset Splitting:** The dataset was split into three distinct subsets:
  - **Training Set:** 909 images
  - **Validation Set:** 196 images
  - **Testing Set:** 195 images
- **Configuration:** A custom data.yaml file was configured to map the paths for training and validation datasets, defining 2 target classes: crop and weed.

## 3. Model Architecture & Training

- **Framework:** Ultralytics YOLOv8 (Nano variant, optimized for edge devices and fast inference).
- **Training Hyperparameters:**
  - **Epochs:** 30
  - **Image Size:** $512 \times 512$ pixels
  - **Batch Size:** 16
  - **Optimizer:** Adam
  - **Environment:** Google Colab GPU (Tesla T4)

## 4. Evaluation & Performance Metrics

The model was evaluated on the validation dataset after 30 epochs of training, achieving strong performance metrics:

- **Precision:** 86.3%
- **Recall:** 77.4%
- **mAP50 (Mean Average Precision at 0.5 IoU):** 86.5%
- **mAP50-95:** 55.6%

**Per-Class Performance Breakdown:**

- **Crop Class:** Precision = 87.3%, Recall = 78.1%, mAP50 = 87.5%
- **Weed Class:** Precision = 85.4%, Recall = 76.8%, mAP50 = 85.4%

## 5. Conclusion

The YOLOv8-based object detection model successfully learned to distinguish between agricultural crops and weeds with high precision and reliability. All project source code, model weights (best_cropweed_model.pt), and configuration files have been successfully uploaded and version-controlled in the public GitHub repository linked above.